

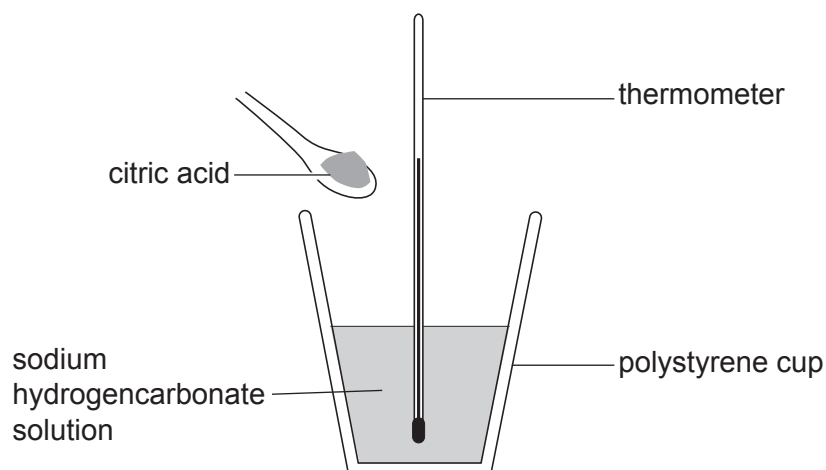
4. (a) Flying Saucers sweets contain sherbet. Sherbet is a mixture of two solid substances – citric acid and sodium hydrogencarbonate.

When put in your mouth, the mixture fizzes and goes cold as it dissolves in your saliva.



A student investigated the temperature change during the reaction between citric acid and a solution of sodium hydrogencarbonate.

Some sodium hydrogencarbonate solution was added to a polystyrene cup and the temperature of the solution was recorded. Citric acid was added to the solution and the temperature was recorded every 10 seconds for 60 seconds.



The results are shown in the table below.

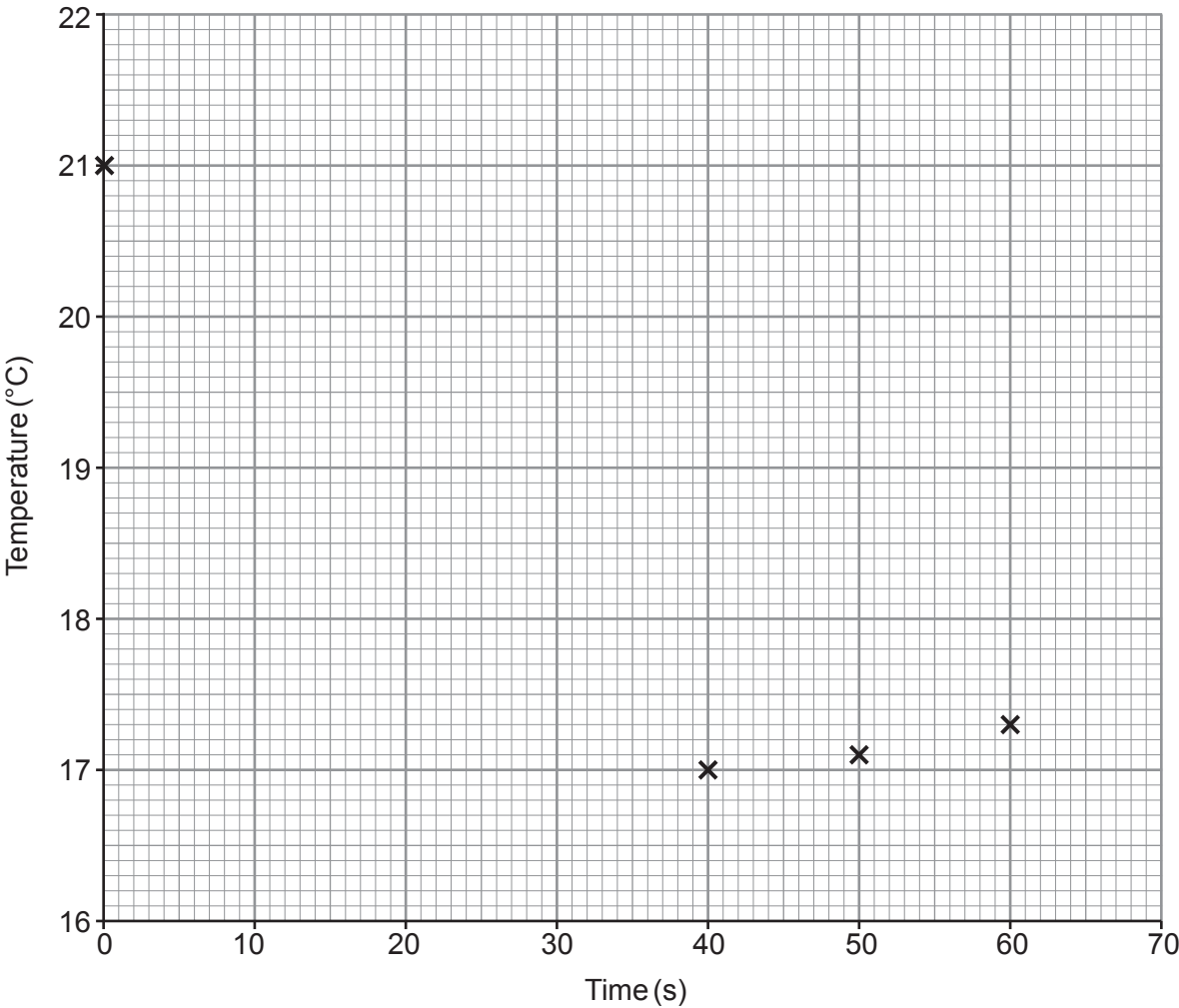
Time (s)	0	10	20	30	40	50	60
Temperature (°C)	21.0	19.0	17.8	17.2	17.0	17.1	17.3

(i) Complete the grid below by plotting the results for 10, 20 and 30 seconds.

The results for 0, 40, 50 and 60 seconds have been plotted for you.

Draw a suitable line using all the results.

[3]



(ii) Use the graph to answer parts I and II.

I. Circle the maximum **change** in temperature during the reaction. [1]

3.7 °C 4.0 °C 21.0 °C 17.3 °C

II. This is described as an endothermic reaction. [1]

Underline the correct word(s) in the brackets to complete the sentence.

In an endothermic reaction the temperature
(**increases** / **stays the same** / **decreases**).

III. The temperature of the contents of the cup was recorded two hours later.

Underline the most likely temperature reading. [1]

4.0 °C 18.0 °C 17.3 °C 21.0 °C

(iii) All the temperature readings were higher than expected due to heat being taken in from the surroundings.

Tick (✓) the box next to the change that would **not** result in the temperature readings being closer to their expected values. [1]

put bubble wrap around the cup

☐

put a lid on the cup

☐

use a cup made from copper

☐

(b) Calculate the relative formula mass (M_r) of sodium hydrogencarbonate, NaHCO_3 . [2]

$$A_r(\text{H}) = 1$$

$$A_r(\text{C}) = 12$$

$$A_r(\text{O}) = 16$$

$$A_r(\text{Na}) = 23$$

$M_r = \dots\dots\dots$

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